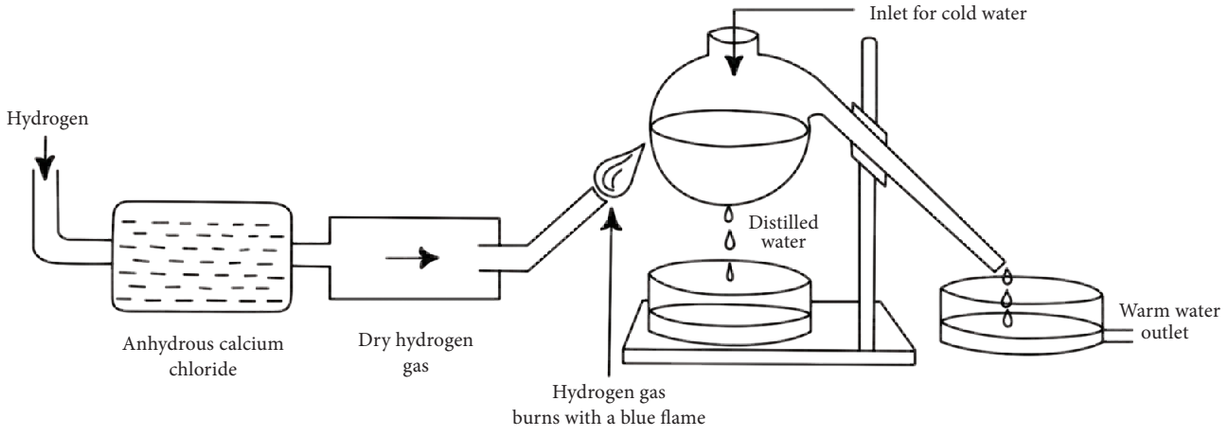


Figure 13.2 Preparation of Water

13.2 Properties of Water

Water has some important properties which are familiar to us. But these properties are unique to water. Some of the physical and chemical properties are explained below.

13.2.1 Physical properties

a. Nature

Pure water is a clear and transparent liquid. It is colourless, odourless and tasteless.

b. Boiling point

The boiling point of water is 100°C at one atmospheric pressure (1 atm). At this temperature, water boils and changes into steam. The boiling point of water increases with increase in pressure. For example, when a pressure cooker is heated, a high pressure is built inside it. The high pressure increases the boiling point of water. Thus, water remains a liquid at a higher temperature (> 100°C) in the cooker. This cooks the food faster.

DO YOU KNOW?

Pure water has the following physical properties.

- Pure water boils at 100° C at one atmospheric pressure.
- Pure water freezes at exactly 0°C at one atmospheric pressure.
- Pure water has a density of 1 gm/cm^3

c. Freezing point

Water freezes at 0°C and forms ice. Thus, the freezing point of water is 0°C. The freezing point of water decreases with increase in pressure.

DO YOU KNOW?

When the skaters move on ice, they exert pressure on it. This pressure lowers the freezing point. As a result, the ice melts underneath the skate and allows the skaters to glide across the ice with little effort. When the skaters move forward, pressure is decreased and the water re-freezes to ice again.



d. Density

When ice cubes are put in a glass of water at room temperature, they float on the surface of the water. This is because ice

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